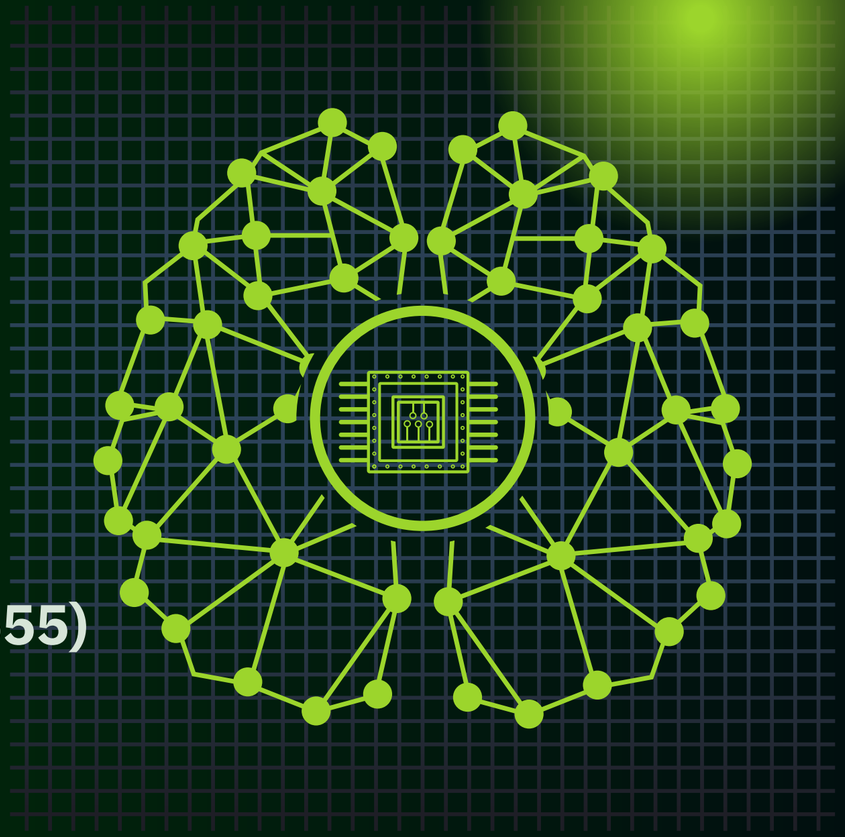
GROUP: TENSORTITANS

DEEP LEARNING PROJECT: WIDS DATATHON 2026

AMIR FARHAN BIN GHAFFAR (2115617)

MUHAMMAD IRSYAD ILHAM BIN AZIZAN (2217555)

AMIR FARHAN BIN GHAFFAR (2115617)



PROJECT DESCRIPTION

When a wildfire ignites, emergency managers face an impossible question with incomplete information. Which fires will reach populated areas? How quickly? And which communities should prepare first?e WiDS Worldwide Global Datathon 2026 is a machine learning competition focused on solving a real-world predictive problem using structured/tabular data. The objective of the competition is to develop a model that can accurately predict a target variable based on a set of input features provided in the dataset.

01 Problem Understanding

This competition requires us to predict a target variable using structured data. It is a supervised learning task where we train a model using labeled data and evaluate its performance on unseen data.

02 Dataset & Evaluation Metric

The dataset includes training and testing data along with metadata. The model is evaluated using AUC-ROC, which measures how well the model distinguishes between classes.

03 EDA & Baseline Model

We performed exploratory data analysis to understand the dataset. Then, we built a baseline deep learning model using a simple neural network to establish initial performance.

04 Results & Conclusion

Our baseline model achieved an AUC score of (0.9012345679012346). While the model is functional, there is room for improvement through better feature engineering and tuning.

1. PROBLEM UNDERSTANDING

The WiDS Worldwide Global Datathon 2026 is a predictive modeling challenge focused on applying machine learning and deep learning techniques to solve a real-world problem. The goal of the competition is to develop a model that can accurately predict the target variable based on the provided dataset.

Key Points

- **Predict a target variable using structured data**
- **Supervised learning problem**
- **Real-world dataset from WiDS Datathon**
- **Focus on:**
  - **Data understanding**
  - **Model development**
  - **Performance evaluation**

Why it matters

- **Helps in real-world decision making**
- **Demonstrates application of deep learning on tabular data**

2. DATASET & EVALUATION METRIC

Dataset Components

- train.csv → features + target
- test.csv → features only
- metaData.csv → feature descriptions
- sample_submission.csv → submission format

Evaluation Metric

AUC-ROC

- Measures classification performance
- Range: 0.5 (random) → 1.0 (perfect)
- Suitable for imbalanced data

3. EDA & BASELINE MODEL

EDA Findings

- Checked missing values
- Analyzed feature distribution
- Examined target balance
- Identified relationships between features

Baseline Model

- Type: Neural Network (MLP)
- Architecture:
  - Input → 64 → 32 → Output
- Activation: ReLU + Sigmoid

Purpose of Baseline

- Provide initial performance
- Ensure model is functional

4. RESULTS & CONCLUSION

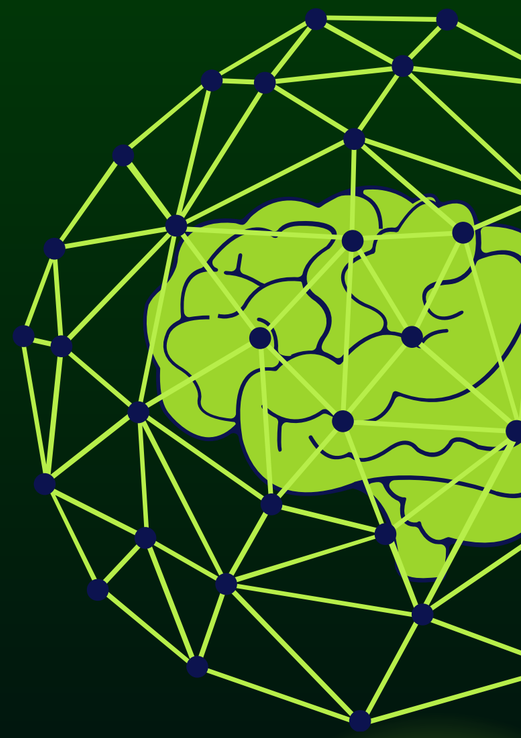
Baseline Performance

- Metric: AUC-ROC
- Score: 0.9012

Key Insights

- Model is functional
- Performance can be improved

Future Improvements

- Feature engineering
- Hyperparameter tuning
- More complex models

THANK YOU